E5 Trigonometric Identities

Name:

Class:

Date:

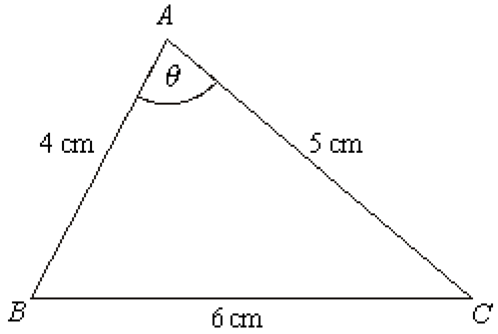
Time: **207 minutes**

Marks: **172 marks**

Comments:

Q1.

The triangle ABC, shown in the diagram, is such that BC = 6 cm, AC = 5 cm and AB = 4 cm. The angle BAC is θ.



(a) Use the cosine rule to show that $\cos\theta = \frac{1}{8}$.

(3)

(b) Hence use a trigonometrical identity to show that $\sin\theta = \frac{3\sqrt{7}}{8}$.

(3)

(c) Hence find the area of the triangle ABC.

(2)

(Total 8 marks)

Q2.

(a) (i) Show that the equation

$2\cot^2 x + 5\operatorname{cosec} x = 10$

can be written in the form $2\operatorname{cosec}^2 x + 5\operatorname{cosec} x - 12 = 0$.

(2)

(ii) Hence show that $\sin x = -\frac{1}{4}$ or $\sin x = \frac{2}{3}$.

(3)

(b) Hence, or otherwise, solve the equation

$2\cot^2(\theta - 0.1) + 5\operatorname{cosec}(\theta - 0.1) = 10$

giving all values of θ in radians to two decimal places in the interval $-\pi < \theta < \pi$.

(3)

(Total 8 marks)

Q3.

(a) Sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$

(3)

(b) Write down the **two** solutions of the equation $\tan x = \tan 61°$ in the interval $0° \leq x \leq 360°$

(2)

(c) (i) Given that $\sin\theta + \cos\theta = 0$, show that $\tan\theta = -1$.

(1)

(ii) Hence solve the equation $\sin(x - 20°) + \cos(x - 20°) = 0$ in the interval $0° \leq x \leq 360°$

(4)

(d) Describe the single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan(x - 20°)$.

(2)

(e) The curve $y = \tan x$ is stretched in the x-direction with scale factor $\frac{1}{4}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(1)

(Total 13 marks)

Q4.

It is given that $2\operatorname{cosec}^2 x = 5 - 5\cot x$.

(a) Show that the equation $2\operatorname{cosec}^2 x = 5 - 5\cot x$ can be written in the form

$$2\cot^2 x + 5\cot x - 3 = 0$$

(2)

(b) Hence show that $\tan x = 2$ or $\tan x = -\frac{1}{3}$.

(2)

(c) Hence, or otherwise, solve the equation $2\operatorname{cosec}^2 x = 5 - 5\cot x$, giving all values of x in radians to one decimal place in the interval $-\pi < x \leq \pi$.

(3)

(Total 7 marks)

Q5.

(a) Describe the single geometrical transformation by which the curve with equation $y = \tan\frac{1}{2}x$ can be obtained from the curve $y = \tan x$.

(2)

(b) Solve the equation $\tan \frac{1}{2} x = 3$ in the interval $\mathbf{0 < x < 4\pi}$, giving your answers in radians to three significant figures.

(4)

(c) Solve the equation

$\cos \theta(\sin \theta - 3 \cos \theta) = 0$

in the interval $0 < \theta < 2\pi$, giving your answers in radians to three significant figures.

(5)

(Total 11 marks)

Q6.

(a) Solve the equation $\sec x = 5$, giving all the values of x in the interval $0 \le x \le 2\pi$ in radians to two decimal places.

(3)

(b) Show that the equation $\tan^2 x = 3 \sec x + 9$ can be written as

$$\sec x^2 - 3 \sec x - 10 = 0$$

(2)

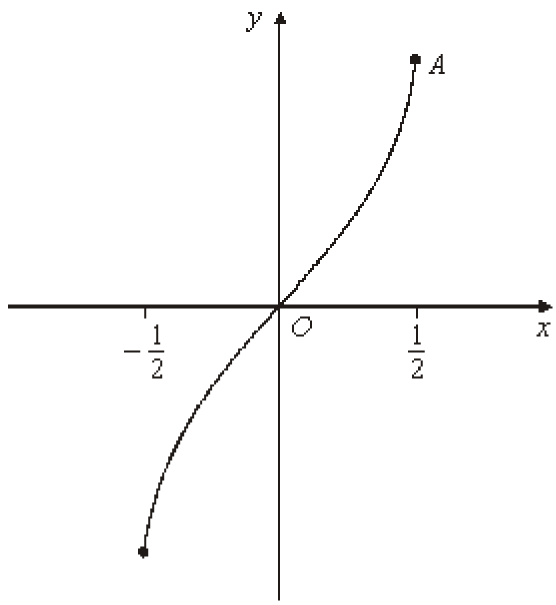
(c) Solve the equation $\tan^2 x = 3 \sec x + 9$, giving all the values of x in the interval $0 \le x \le 2\pi$ in radians to two decimal places.

(4)

(Total 9 marks)

Q7.

The diagram shows the curve with equation $y = \sin^{-1} 2x$, where $-\frac{1}{2} \le x \le \frac{1}{2}$.



(a) Find the y-coordinate of the point A, where $x = \frac{1}{2}$.

(1)

(b) (i) Given that $y = \sin^{-1} 2x$, show that $x = \frac{1}{2}\sin y$.

(1)

(ii) Given that $x = \frac{1}{2}\sin y$, find $\frac{dx}{dy}$ in terms of y.

(1)

(c) Using the answers to part (b) and a suitable trigonometrical identity, show that

$$\frac{dx}{dy} = \frac{2}{\sqrt{1-4x^2}}$$

(4)

(Total 7 marks)

Q8.

(a) (i) Show that the equation

$4\tan\theta\sin\theta = 15$

can be written as

$4\sin^2\theta = 15\cos\theta$

(1)

(ii) Use an appropriate identity to show that the equation

$$4 \sin^2\theta = 15 \cos \theta$$

can be written as

$$4 \cos^2\theta + 15 \cos \theta - 4 = 0$$

(2)

(b) (i) Solve the equation $4c^2 + 15c - 4 = 0$.

(2)

(ii) Hence explain why the only value of $\cos \theta$ which satisfies the equation

$$4 \cos^2\theta + 15 \cos\theta - 4 = 0$$

is $\cos \theta = \frac{1}{4}$.

(1)

(iii) Hence solve the equation $4 \tan\theta \sin\theta = 15$ giving all solutions to the nearest 0.1° in the interval $0° \leq \theta \leq 360°$.

(2)

(c) **Write down** all the values of x in the interval $0° \leq x \leq 90°$ for which

$$4 \tan 4x \sin 4x = 15$$

giving your answers to the nearest degree.

(2)

(Total 10 marks)

Q9.

It is given that $\tan^2 x = \sec x + 11$.

(a) Show that the equation $\tan^2 x = \sec x + 11$ can be written in the form

$$\sec^2 x - \sec x - 12 = 0$$

(2)

(b) Hence show that $\cos x = \frac{1}{4}$ or $\cos x = -\frac{1}{3}$.

(3)

(c) Hence, or otherwise, solve the equation $\tan^2 x = \sec x + 11$, giving all values of x to the nearest degree in the interval $0° < x < 360°$.

(3)

(Total 8 marks)

Q10.

(In this question, take g = 10 m s^{-2}.)

A projectile is fired from a point O with speed u at an angle of elevation α above the horizontal so as to pass through a point P. The projectile travels in a vertical plane through O and P. The point P is at a horizontal distance $2k$ from O and at a vertical distance k above O.

(a) Show that α satisfies the equation

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0$$

(7)

(b) Deduce that

$$u^4 - 20ku^2 - 400k^2 \geq 0$$

(3)

(Total 10 marks)

Q11.

(In this question, use g = 10 m s^{-2}.)

A golf ball is hit from a point O on a horizontal golf course with a velocity of 40 m s^{-1} at an angle of elevation θ. The golf ball travels in a vertical plane through O. During its flight, the horizontal and upward vertical distances of the golf ball from O are x and y metres respectively.

(a) Show that the equation of the trajectory of the golf ball during its flight is given by

$$x^2 \tan^2\theta - 320x \tan\theta + (x^2 + 320y) = 0$$

(6)

(b) (i) The golf ball hits the top of a tree, which has a vertical height of 8 m and is at a horizontal distance of 150 m from O.

Find the two possible values of θ.

(5)

(ii) Which value of θ gives the shortest possible time for the golf ball to travel from O to the top of the tree? Give a reason for your choice of θ.

(2)

(Total 13 marks)

Q12.

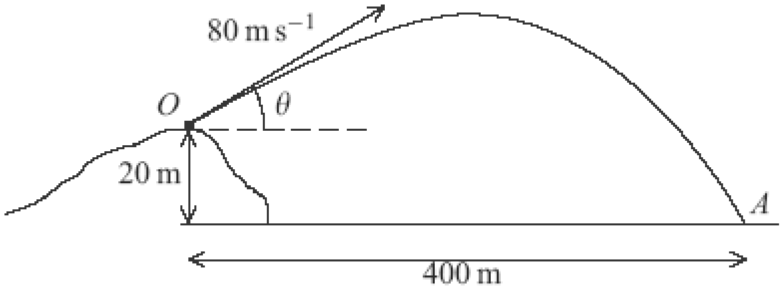
A projectile is fired from a point O on top of a hill with initial velocity 80 m s^{-1} at an angle θ above the horizontal and moves in a vertical plane. The horizontal and upward vertical distances of the projectile from O are x metres and y metres respectively.

(a) (i) Show that, during the flight, the equation of the trajectory of the projectile is given by

$$y = x\tan\theta - \frac{gx^2}{12\,800}\left(1 + \tan^2\theta\right)$$

(5)

(ii) The projectile hits a target A, which is 20 m vertically below O and 400 m horizontally from O.



Show that

$$49\tan^2\theta - 160\tan\theta + 41 = 0$$

(2)

(b) (i) Find the two possible values of θ. Give your answers to the nearest 0.1°.

(3)

(ii) Hence find the shortest possible time of the flight of the projectile from O to A.

(2)

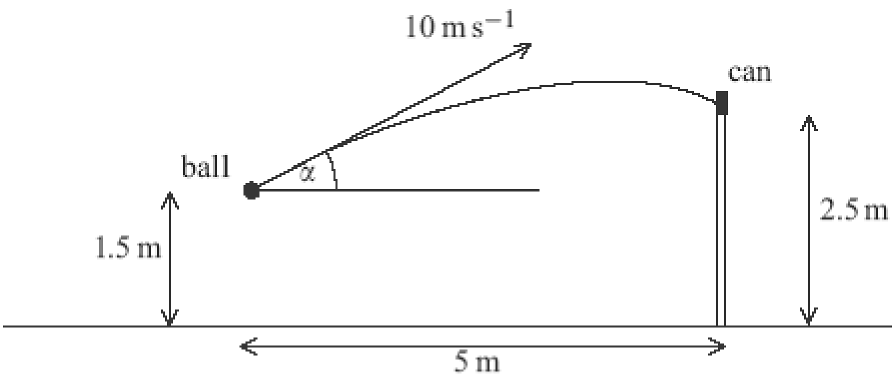
(c) State a necessary modelling assumption for answering part (a) (i).

(1)

(Total 13 marks)

Q13.

A boy throws a small ball from a height of 1.5 m above horizontal ground with initial velocity 10 m s^{-1} at an angle α above the horizontal. The ball hits a small can placed on a vertical wall of height 2.5 m, which is at a horizontal distance of 5 m from the initial position of the ball, as shown in the diagram.



(a) Show that α satisfies the equation

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

(7)

(b) Find the two possible values of α, giving your answers to the nearest 0.1°.

(3)

(c) (i) To knock the can off the wall, the horizontal component of the velocity of the ball must be greater than 8 m s^{-1}.

Show that, for one of the possible values of α found in part (b), the can will be knocked off the wall, and for the other, it will **not** be knocked off the wall.

(3)

(ii) Given that the can is knocked off the wall, find the direction in which the ball is moving as it hits the can.

(4)

(Total 17 marks)

Q14.

A ball is projected with speed u m s^{-1} at an angle of elevation a above the horizontal so as to hit a point P on a wall. The ball travels in a vertical plane through the point of projection. During the motion, the horizontal and upward vertical displacements of the ball from the point of projection are x metres and y metres respectively.

(a) Show that, during the flight, the equation of the trajectory of the ball is given by

$$y = x \tan \alpha - \frac{gx^2}{2u^2}\left(1 + \tan^2 \alpha\right)$$

(6)

(b) The ball is projected from a point 1 metre vertically below and R metres horizontally from the point P.

(i) By taking g = 10 m s^{-2}, show that R satisfies the equation

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

(2)

(ii) Hence, given that u and R are constants, show that, for tan a to have real values, R must satisfy the inequality

$$R^2 \leq \frac{u^2\left(u^2 - 20\right)}{100}$$

(2)

(iii) Given that R = 5, determine the minimum possible speed of projection.

(3)

(Total 13 marks)

Q15.

A football is kicked from a point O on a horizontal football ground with a velocity of 20 m s^{-1} at an angle of elevation of 30°. During the motion, the horizontal and upward vertical displacements of the football from O are x metres and y metres respectively.

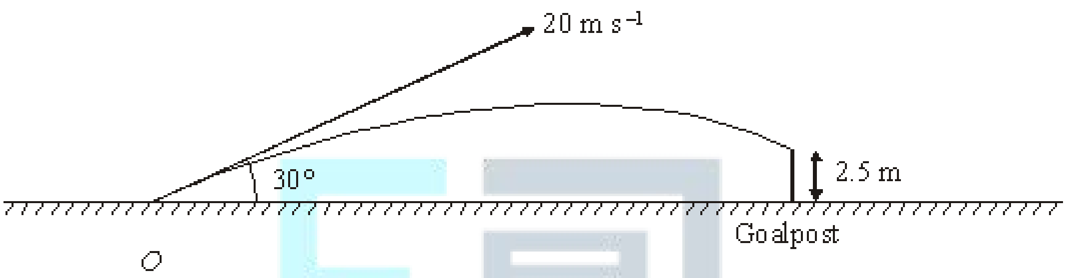
(a) Show that x and y satisfy the equation

$$y = x \tan 30° - \frac{gx^2}{800 \cos^2 30°}$$

(6)

(b) On its downward flight the ball hits the horizontal crossbar of the goal at a point which is 2.5 m above the ground. Using the equation given in part (a), find the horizontal distance from O to the goal.

(4)



(c) State **two** modelling assumptions that you have made.

(2)
(Total 12 marks)

Q16.

A golf ball is driven with speed 40 m s^{-1} at an angle α to the horizontal from a point on a horizontal golf course. The ball travels in a vertical plane.

(a) Show that, during its flight, the horizontal and vertical displacements, x metres and y metres respectively, of the ball from the point of projection satisfy the equation

$$y = x \tan \alpha - \frac{gx^2}{3200}(1 + \tan^2 \alpha)$$

(5)

(b) The golf ball just clears a tree of vertical height 4 m at a horizontal distance of 100 m from its point of projection.

Find the two possible values of α.

(6)

(c) State two modelling assumptions that you have made.

(2)
(Total 13 marks)